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SEEP: Semantic-enhanced question embeddings pre-training for improving knowledge tracing

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ABSTRACT

Knowledge Tracing (KT) defines the task of diagnosing students' dynamic knowledge level in exercises. Although existing efforts have leveraged question information, most of them either learn question embeddings during the process of model training or represent questions based on the correlation between questions and concepts, which ignores plentiful implicit information entailed in the student-question-concept interaction and the revelation of fine-grained semantics between the interaction as well as the usage of the students' historical answers. It is, however, challenging to extract and refine implicit information in the student-question-concept interaction which is highly heterogeneous and complex. To this end, in this paper, we present a novel **S**emantic-**E**nhanced Question **E**mbeddings **P**re-training (**SEEP**) method, concentrating on decomposing underlying relation information in the interaction and further fusing information of questions and concepts under different decomposed semantic perspectives to obtain semantic-enhanced question embeddings for improving performances of KT methods. Extensive experiments conducted on two real-world datasets show SEEP has the higher expressive power that enables KT methods to predict students' performance.

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1. Introduction

The recent COVID-19 epidemic has triggered the outbreak and proliferation of online education platforms which create an open as well as convenient environment for both students and teachers to learn and teach at home [1,2]. Now more than ever, it has become paramount to push the limits of online learning in every manner to keep flourishing the education system, where an essential component of online education is knowledge tracing (KT) aiming at modeling student's knowledge proficiency on the basis of their answers to a sequence of questions referred as interactions or responses.

The purpose of KT is predicting the student's response score on next candidate question. In view of the goal of KT, great efforts have been made have achieved impressive results on student performance prediction. For example, BKT [3] and PFA [4] are two traditional methods and performs well. Recently, many outstanding deep learning-based works are proposed to solve the knowledge tracing task, and basically, these methods can be divided into three broad strands of works: memory-based methods, transformer-based methods and graph-based methods. First, memory-based methods like DKT [5], DKVMN [6] and LPKT [7] try to learn sequential patterns from individual student response logs by using temporal models. Usually,

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concepts in these temporal models are treated as latent proficiency representations, such that performance predictions can be then generated based on these representations. Second, due to the characteristics of transformer framework, some transformer-based methods including SAKT [8], RKT [9], and AKT [10] are proposed and improve performances of those memory-based methods. Third, unlike memory-based and attention-based methods, graph-based methods excavate the interaction between questions and concepts with certain graphs, such as GKT [11] and GIKT [12].

Although existing methods utilized question information for KT, their flexibility and comprehensiveness were insufficient yet. To be more specific, there exists two problems. First, existing methods typically learned the question embeddings during the process of training, where the question embeddings were deeply entangled with the model and fail to generalize on other KT model architectures. Second, in addition to flexibility, most of these methods represented a question based on correlations between questions and concepts, neglecting rich relation information contained in the student-question-concept interaction, the revelation of finer-grained semantics for interaction and the usage of the students' response logs.

To motivate this work, we conduct a statistical analysis on ASSISTment09 which is an example pre-processed dataset (Table 1 shows its statistics, and more details will be specified in Section 5). Our intention is to validate that there exists a substantial amount of relation information underexplored. More specifically, we present two different distributions of number of questions, as reported in Fig. 1. Depending on Table 1 and Fig. 1, we can observe that: (1) a question may involve one or more concepts (shown in Fig. 1)) and most of concepts are relevant to adequate questions (shown in Fig. 1(b)), which implies that there exists a large number of underlying co-occurrence patterns between question pairs and concept pairs; (2) from the perspective of answer-agreement, sufficient response records are provided in ASSISTment09 as shown in Table 1, which hints at the potential of revealing fine-grained semantics between the interaction and the students' response records. Therefore, to generate question embeddings, exploiting question pairs and concept pairs comprehensively from two different semantic perspective (i.e., co-occurrence and answer-agreement) is a well-motivated problem and of great necessity.

As previous studies have demonstrated, learning informative question embeddings by considering the student-question-concept interaction is more comprehensive than the question-concept interaction. Although great efforts have been made, rich information contained in the student-question-concept interaction is not effectively revealed and utilized yet. Furthermore, in our early study, we have found that response data can reflect question difficulty information to some extent, which is conducive to pre-training informative question embeddings. Along this line, in this paper, we attempt to focus on decomposing latent information in the interaction and further fusing the information of questions and concepts under different discomposed relations to attain semantic-enhanced question embeddings for improving the performance of KT models.

However, difficulties lie ahead to obtain ideal question embeddings owing to the complexity of the student-question-concept interaction. On one hand, the graph of student-question-concept interaction contains different types of nodes and edges that make it highly heterogeneous as illustrated in Fig. 2. On the other hand, the structure of the student-question-concept interaction graph is pretty complex and its hierarchy is also directly explicit, which indicates that modeling question embeddings by exploiting the student-question-concept interaction graph comprehensively is very challenging. To remedy the issues of modeling the co-existing heterogeneity and comprehensiveness, we propose a general question embeddings pre-training method, namely Semantic-Enhanced Question Embeddings Pre-training (SEEP) which has the ability of generalizing on other KT methods and takes the plentiful relation information entailed in the student-question-concept interaction into consideration. Hinged upon the above insights, this method focuses on decomposing implicit information in the interaction and further fusing the information of questions and concepts under different discomposed semantics to attain semantic-enhanced question embeddings for improving the performance of KT models. Concretely, from two semantic perspectives (co-occurrence and answer-agreement), we first mine relations between questions (and concepts) as the prior. Thereafter, by extracting question-concept bipartite graph from the student-question-concept interaction, we utilize a two-level attention aggregation mechanism with the prior information to attain node embeddings of questions and concepts. Furthermore, for obtaining the final question embeddings, we solve a joint optimization problem by taking question difficulty constraint and question-concept relation structure constraint into account. Finally, extensive experiments conducted on two real-world datasets clearly demonstrate that SEEP has higher expressive power that enables KT methods to effectively predict students' performance. In summary, our key contributions are listed as follows:

1. From two decomposed perspectives of co-occurrence and answer-agreement, we strive to exploit the relations between questions (and concepts) as prior information;
2. To comprehensively model question embeddings, we design a two-level attention network to organically combine questions and concepts with the prior and side information of questions;
3. Extensive experiments on two real-world datasets validate the effectiveness and superiority of SEEP, including quantitative comparison, qualitative analysis and case study.

The rest of this paper is organized as follows: we briefly review the related work in Section 2. Some preliminaries and important notations are specified in Section 3. In Section 4, the proposed SEEP method is presented and discussed in detail. We conduct extensive experiments on two real-world datasets and comprehensively explore these results in Section 5. Section 6 concludes the paper and identifies future work which is likely to be carried out on the basis of the new findings exhibited in this paper.

Table 1
The statistics of ASSISTment09.

Statistics			
#Students	4,003	#Concepts per question	2.17
#Questions	3,215	#Std of concepts per question	0.47
#Concepts	123	#Questions per concept	172.69
#Response records	50,551	#Std of questions per concept	169.58

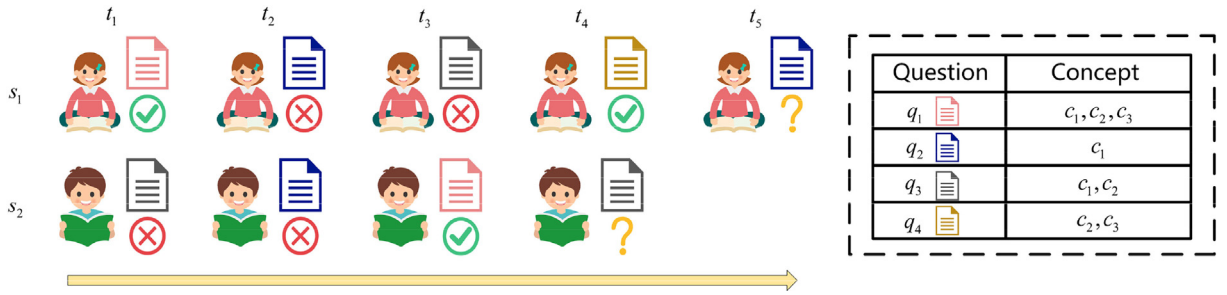


Fig. 1. A statistical analysis on ASSISTment09.

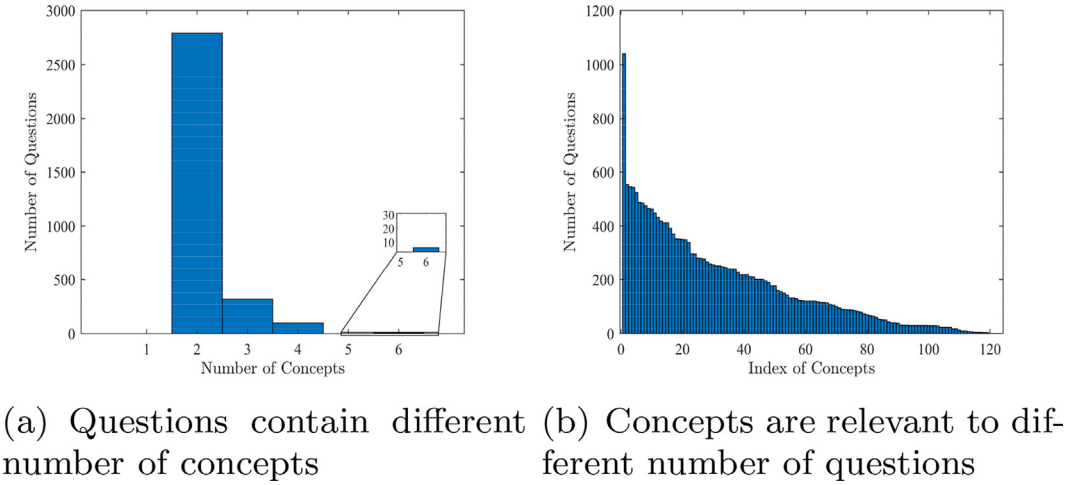


Fig. 2. The illustrative examples of (a) the student-question-concept interaction; (b) the heterogeneous graph of student-question-concept interaction, and please note that edges representing student-question interactions do not reveal response data (i.e., answer correctly or not).

2. Related work

Our proposed method is pertinent to two broad strands of the research literature, including knowledge tracing and question embedding, respectively.

2.1. Knowledge tracing

The developed method is designed for knowledge tracing, a fundamental task, which is indirectly inferring the probability of students' success/failure on a candidate question and evaluating the students' knowledge proficiency of concepts on the basis of sequences of their past attempts [13]. Most of the existing knowledge tracing models can be classified into traditional probabilistic models, logistic models, and deep learning-based methods. Bayesian Knowledge Tracing (BKT) [3] is a classic and widely-used probabilistic model for KT, which can be seen as a special case of the Hidden Markov Model (HMM). Logistic models are a large class of models based on logistic functions, which utilizes a logistic function to estimate the probability of knowledge state, such as Performance Factor Analysis (PFA) [4]. Following the idea of mainstream KT methods, these methods can be largely categorized into three types: memory-based methods, transformer-based methods and graph-based methods. Specifically, first, recurrent neural networks (RNNs) are well-designed for sequential tasks so that

it can be naturally applied on KT task. For example, DKT [5], DKVMN [6] and LPKT [7] are presented to model sequential interactions between students and questions. Since these memory-based methods are notoriously unable to capture dependency relationships between sequential interactions and hard to parallel, some transformer-based methods including SAKT [8], RKT [9], and AKT [10] are proposed and outperform the memory-based methods. Although transformer-based methods take dependency relationships between interactions into account, they still fail to regard the student-question-concept interaction as a heterogeneous graph and ignore the significance of plentiful relation information excavation existed in such a heterogeneous graph. For this, GKT [11] and GIKT [12] are developed, where GKT builds a similarity graph of concepts randomly and automatically learns the edge weights of the graph explicitly while GIKT proposes an end-to-end framework to employ high-order question-concept relation graphs into question and concept representations.

Although these aforementioned methods have made great progress, none of these question embeddings of them is of flexibility enough so that those learned question embeddings are unable to generalize on other KT model architectures. Along this line, we attempt to develop a pre-trained question embedding method with abundant semantic information for downstream KT methods.

2.2. Question embedding

Another stream of related work is question embedding [14–16]. Generally, question embedding methods can be divided into two categories: model-dependent methods, where the question embeddings are deeply entangled with the model, and model-free methods, where parameters of models are pre-trained on corpus and then fine-tuned in specific educational tasks. More specifically, as for model-dependent methods, it can broadly fall into three classes. First, analogous to DKT and DKVMN, this family of methods usually only leverages the index information like one-hot vectors to represent questions. Unfortunately, such a field-wise one-hot encoding representation is prone to curse of dimensionality and enormous sparsity. Moreover, some advanced methods are proposed using side information, especially text information, to enhance the representative ability of question embedding, such as RKT [9] and EKT [17], whereas the text descriptions are not easy to acquire in practice. Considering the limitation of text information of questions, most of model-dependent methods like NeuralCD [18], RCD [19], SAKT [20], and AKT [21], try to exploit semantic information from training datasets as fully as possible to automatically learn question embeddings, but the question embeddings are deeply entangled with the model.

To make question embeddings more flexible without weakening their representative ability, another line of question embedding methods aims to be model-free. QuesNet [14] is the first attempt to pre-train the representations of educational questions from the heterogeneous data. In addition, DisenQNet [15] is a novel unsupervised model, which aims to distinguish the diverse characteristics of questions and learn the disentangled representations of questions. However, both of these two model-free methods encode text descriptions of questions into question embeddings to capture the question characteristics. In order to get rid of the limitation of texts, inspired by LINE [20], PEBG [16] is designed and takes a further step towards maximally extracting and exploiting plentiful underlying information among questions and concepts. Although PEBG model question embeddings from question-concept interaction graph without text information, it overlooks rich relation information contained in the student-question-concept interaction graph, thus leaving abundant information underexploited, such as fine-grained semantics between the interaction and the students' response logs. PERM [22] utilizes the relation map to pre-train question embeddings for improving knowledge tracing, while it ignores the inherent question difficulty information.

Therefore, in this paper, we strive to model question embeddings which have the ability of generalizing on downstream KT methods and are able to take the relation information contained in the student-question-concept interaction graph into consideration.

3. Preliminaries

In this section, we first present a toy example to motivate this paper. Then, we introduce some preliminary knowledge about question embedding, including necessary definitions and notations used in this paper and then present an overview for our framework.

3.1. Motivational example

As illustrated in Fig. 3, the dotted box states the correlations between questions and concepts. Specifically, taking student s_1 in Fig. 3 as an example, first some items from a set of questions (e.g., q_1, q_2, q_3 and q_4) are assigned to this girl to practice and she leaves her response logs (e.g., right or wrong) at different timestamps. For concepts, the pattern (c_1, c_2) appears in q_1 and q_3 while the pattern (c_1, c_3) only exists in q_1, which means c_1 is more relevant to c_2 than c_3 from the semantic of co-occurrence. Besides, both q_2 (containing c_1) and q_3 (containing c_1 and c_2) are incorrectly answered twice (i.e., c_1 is answered incorrectly four times and c_2 is answered incorrectly twice) by these two students, which may indicate that there exists a certain relation between c_1 and c_2 from the perspective of answer-agreement. Thus, along this line, an ideal question embedding method should be flexible and ought to exploit the student-question-concept interaction comprehensively.

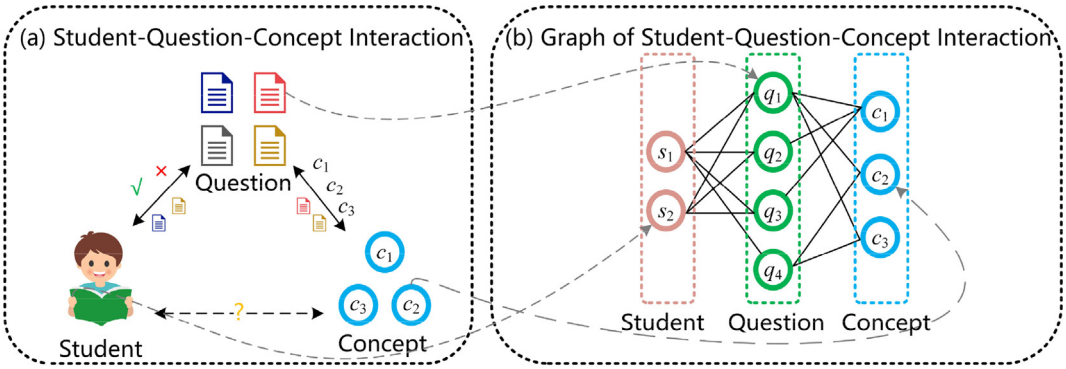


Fig. 3. A toy example of students' learning process. There are two students who are assigned some questions to practice and leave their answers (correct or wrong) at different timesteps.

3.2. Problem formulation

Let $S = \{s_1, s_2, \ldots, s_N\}, Q = \{q_1, q_2, \ldots q_M\}$ and $C = \{c_1, c_2, \ldots, c_K\}$, where N is the number of students, M is the number of questions, and K is the number of knowledge concepts at a learning system, respectively. In KT task, a student selects some questions from Q for practice, and his/her response logs $X = \{x_1, x_2, \ldots, x_T\}$ are characterized by a set of tuple, where $x_t = (q_t, a_t)$ denotes the question attempted, $a_t \in \{0, 1\}$ denotes the correctness of the corresponding answer which is left by a student at timestep t. Usually, a student may answer more than one question, and one question is related to several concepts, and vice versa. In this way, the student-question-concept interaction can be represented as a heterogeneous graph $G = (V, R)$, where $V = S \cup Q \cup C$ and $R = \{R_{sq}, R_{qc}\}$. For clarity, here we do not reveal students' response data (i.e., right or wrong) and their attempt times. Next, we introduce some definitions we will leverage to train question embeddings in our model, including the graph information, the transaction information and the difficulty information.**Definition 1 (question-concept bipartite graph)**. The question-concept bipartite graph, denoted as $G_{qc} = (Q \cup C, R_{qc})$, is an undirected graph, where $R_{qc} = \{r_{q_i \leftrightarrow c_k} | q_i \in Q, c_k \in C, r_{q_i \leftrightarrow c_k} \in \{0, 1\}\}$. In addition, $r_{q_i \leftrightarrow c_k} = 1$, refers to that question q_i includes the concept c_k, otherwise $r_{q_i \leftrightarrow c_k} = 0$.

Definition 2 (question transaction). A question transaction refers to a concept c_k, denoted as $tr_{c_k} = \{q_i | r_{q_i \leftrightarrow c_k} = 1\}$.

Definition 3 (concept transaction). A concept transaction refers to a question q_i, denoted as $tr_{q_i} = \{c_k | r_{q_i \leftrightarrow c_k} = 1\}$.

Definition 4 (question difficulty). Following previous works [16,23], the question difficulty d_i of a certain question q_i is defined as the ratio of incorrectly answered by computing from the training dataset.

3.3. Framework overview

Fig. 4 illustrates an overview of the proposed Semantic-Enhanced Question Embeddings Pre-training (SEEP) framework, consisting of four layers: input layer, decomposition layer, aggregation layer and constraint layer, respectively. Concretely, the input layer is devised to model the student-question-concept interaction as a heterogeneous graph uniformly. Furthermore, due to the issue of co-existing *heterogeneity*, relation information existed in the heterogeneous graph is decomposed into two semantic perspectives (i.e., co-occurrence and answer-agreement) by the well-designed decomposition layer. Afterwards, to tackle the challenge of *comprehensiveness*, we propose the aggregation layer to attain the node embeddings of questions and concepts depending on a two-level attention aggregation mechanism with the prior information. Finally, to preserve the difficulty and structure information of question embeddings, the constraint layer is developed to predict the difficult degree of the target question and to estimate the local proximity between questions and concepts. For better description, the key notations are summarized in Table 2.

4. Method

In this section, we propose a novel question embedding pre-training method (SEEP) for knowledge tracing. The SEEP method is motivated by four intuitions rooted in intelligent education [14–16,24] and heterogeneous graph embedding [25,26], and we will detail these intuitions as follows.

4.1. Input layer

Before pre-training question embeddings for knowledge tracing, we should notice that questions exist in the student-question-concept interaction. Moreover, each object (i.e., student, question or concept) in the interaction plays

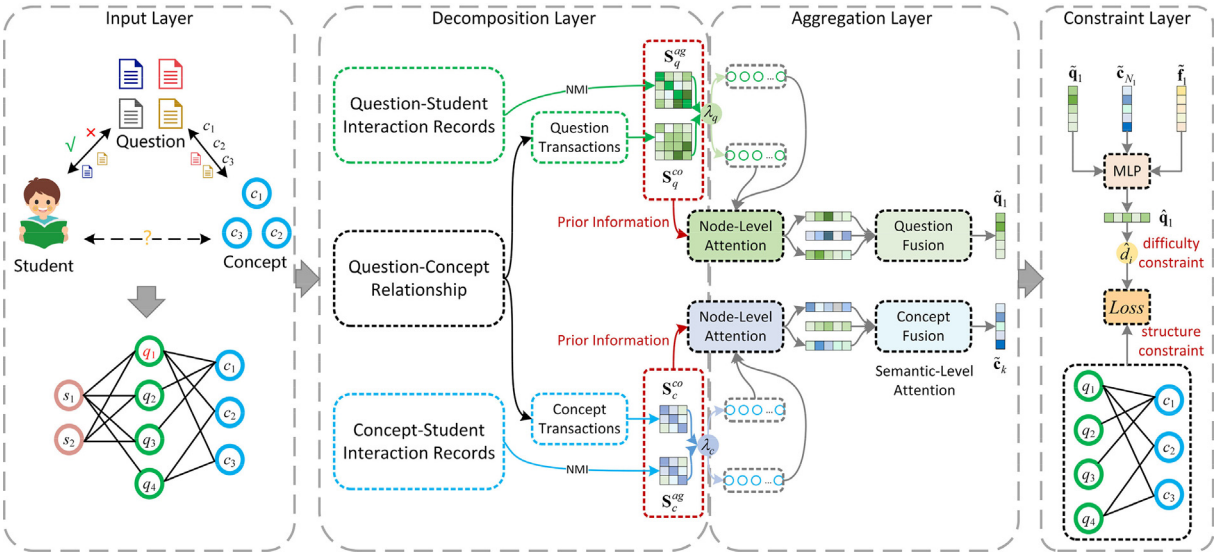


Fig. 4. The SEEP framework overview.

Table 2
The key mathematical notations and descriptions.

Notation	Description
N, M, K	the total number of students; the total number of exercises; the total number of concepts
tr_{c_k}, tr_{q_i}	a question transaction; a concept transaction
$d_i, \hat{d}_i$	the actual difficulty of question i; the predicted difficulty of question i
$\mathbf{q}_i, \mathbf{c}_k$	the initial embedding of question i; the initial embedding of concept k
$\tilde{\mathbf{q}}_i, \tilde{\mathbf{c}}_k, \hat{\mathbf{q}}_i$	the updated embedding of question i; the updated embedding of concept k; the final embedding of question i
$\mathbf{T}^q$	the question transaction matrix
$\mathbf{S}_q^{co}, \mathbf{S}_q^{ag}$	the question relation matrix under the co-occurrence perspective; the question relation matrix under the answer-agreement perspective
$\widehat{\mathbf{S}}_q^{co}, \widehat{\mathbf{S}}_q^{ag}$	the normalized question relation matrix under the co-occurrence perspective; the normalized question relation matrix under the answer-agreement perspective
ψ, ϕ	a certain semantic perspective $\psi \in \{co, ag\}$; a certain semantic perspective $\phi \in \{co, ag, dir\}$
$\mathbf{e}_{q_i}^{\psi}, \mathbf{e}_{c_k}^{\psi}$	the learned embedding of question q_i for the perspective ψ; the learned embedding of concept c_k for the perspective ϕ
$\lambda_q, \lambda_c, \gamma$	the sparsity factor of question q; the sparsity factor of concept k; the balance factor

a different role and shows different significance in learning their embeddings for the specific task. This observation encourages our first intuition: *student-question-concept interaction can be naturally modeled as a heterogeneous graph, where each node represents an object in the interaction and edges exist between nodes presenting explicit semantic signals.* Although students, questions and concepts are able to be represented as nodes in the heterogeneous graph, for node embedding, we solely focus on learning question embeddings and concept embeddings. As relation information (i.e., answer correctly or incorrectly) hidden in the interactions of student-question and student-concept is also heterogeneous, it is inherently difficult to be modeled as different types of edges and a solution to this problem will be presented in Section 4.2.

Thereby, in the input layer, we encode questions and concepts with d-dimensional trainable matrices $\mathbf{Q} \in \mathbb{R}^{M \times d}$ and $\mathbf{C} \in \mathbb{R}^{K \times d}$. For question q_i, it is initialized by multiplying its one-hot representation vector $\mathbf{x}_i$ with the trainable matrix $\mathbf{Q}$ (i.e., the transpose of i-th row from matrix $\mathbf{Q}$). Similarly, concept c_k's initialized embedding is the transpose of k-th row from matrix $\mathbf{C}$. Specifically, we model the input layer as:

$$\begin{cases} \mathbf{q}_i = \mathbf{x}_i^T \mathbf{Q} \\ \mathbf{c}_k = \mathbf{x}_k^T \mathbf{C} \end{cases} \tag{1}$$

where $\mathbf{q}_i, \mathbf{c}_k \in \mathbb{R}^d, \mathbf{x}_i \in \{0,1\}^M$ and $\mathbf{x}_k \in \{0,1\}^K$.

4.2. Decomposition layer

As the graph of student-question-concept interaction is highly heterogeneous and leaves implicit information underexplored, we attempt to decompose different kinds of relation information from different perspectives. Thus, technically, a new decomposition layer is designed, which reflects our second intuition: *numerous implicit information is deeply entangled with the heterogeneous graph so that decomposing these information explicitly from the graph is relatively rational and desired*. To be more specific, the decomposition layer is developed to mine those underlying relation information from co-occurrence perspective and answer-agreement perspective, respectively.

4.2.1. Co-occurrence perspective

In terms of co-occurrence perspective, concentrating on the question-concept bipartite graph, we are dedicated to finding homogeneous neighbors of questions and concepts. Particularly, following the definitions of R_{qc} and tr_{c_k} rooted in [27], the question transaction matrix can be specified as $\mathbf{T}^q = [t_{ij}] \in \left\{ r_{c_i \leftrightarrow q_j} \right\}^{K \times M}$, and each row of this matrix can be regarded as a question transaction (i.e., tr_{c_k}). Therefore, question relation matrix under the co-occurrence perspective is denoted as $\mathbf{S}_q^{co}$, and one of its elements $S_q^{co}[ij]$ indicates the relation strength between q_i and q_j. Specifically, the strength of relation between any two questions can be calculated as Eq. (2).

$$S_q^{co}[ij] = \frac{\frac{\left\| \mathbf{T}^q_{:,i} \circ \mathbf{T}^q_{:,j} \right\|_1}{K}}{\frac{\left\| \mathbf{T}^q_{:,i} \right\|_1}{K} \times \frac{\left\| \mathbf{T}^q_{:,j} \right\|_1}{K}} = \frac{\left\| \mathbf{T}^q_{:,i} \circ \mathbf{T}^q_{:,j} \right\|_1}{\left\| \mathbf{T}^q_{:,i} \right\|_1 \times \left\| \mathbf{T}^q_{:,j} \right\|_1 / K} \tag{2}$$

where $\mathbf{T}^q_{:,i}$, $\mathbf{T}^q_{:,j}$ are the i-th column of the matrix $\mathbf{T}^q$ and the j-th column of the matrix $\mathbf{T}^q$ respectively, $\circ$ is the element-wise product, and $\|\cdot\|_1$ is the 1-norm of a vector. The value of $S_q^{co}[ij]$ lies in $[0, +\infty)$ according to the property of Eq. (2). Furthermore, in view of conditional probability, if $S_q^{co}[ij] = 1$, q_i and q_j follow an independent distribution, which means that there is no explicit relation between q_i and q_j. If $S_q^{co}[ij] > 1$ ($S_q^{co}[ij] < 1$), there is a positive (negative) relation between q_i and q_j.

Afterwards, to obtain homogeneous neighbors of the target question q_i, we define the normalized question relation matrix under the co-occurrence perspective as $\widehat{\mathbf{S}}_q^{co} \in \{0,1\}^{M \times M}$:

$$\widehat{S}_q^{co}[ij] = \begin{cases} 1, \text{if} \left(S_q^{co}[ij] > 1 \right) \wedge \left(\left(S_q^{co}[ij] / \max\left(\mathbf{S}_q^{co} \right) \right) \geqslant \lambda_q \right) \\ 0, \text{otherwise} \end{cases} \tag{3}$$

where λ_q is a hyperparameter to control the sparsity of $\widehat{\mathbf{S}}_q^{co}$. Hence, the set of homogeneous neighbors $N_{q_i}^{co} = \left\{ q_j | \widehat{S}_q^{co}[ij] = 1, q_j \in Q \right\}$ is readily available for the target question q_i. Along this line, the set of homogeneous neighbors $N_{c_k}^{co}$ for a certain concept c_k can be also obtained on the basis of hyperparameter λ_c.

4.2.2. Answer-agreement perspective

For answer-agreement perspective, we also strive to obtain homogeneous neighbors of questions and concepts. As mentioned before, relation information (i.e., answer correctly or incorrectly) hidden in the interactions of student-question and student-concept is inherently difficult to be modeled in the heterogeneous graph as shown in Fig. 3(b). Thus, we will now describe how students' performance data can be used to obtain the relevance between questions and the relevance between concepts. First, by solely considering the pairs of q_i and q_j, we can find all the question pairs (q_i, q_j) in each student's interaction sequence, where q_i and q_j should occur concurrently in a response sequence as illustrated in Fig. 1. Then, we compute the *NMI* (Normalized Mutual Information) which is popularly used as a measure of association for two variables [28] to represent the strength of relation $S_q^{ag}[ij]$. We view the response of q_i and q_j (i.e., a_i and a_j) as two random variables respectively, mathematically $S_q^{ag}[ij]$ can be calculated as Eq. (4). Concretely, probabilities in the Eq. (4) can be derived by counting frequencies within all the satisfied interaction pairs, where m and n are different answers (either 1 or 0), representing different values of random variables (i.e., a_i and a_j). Actually, $S_q^{ag}[ij]$ reflects the degree of dependency between q_i and q_j. Similarly, to acquire homogeneous neighbors of the target question q_i, we denote the normalized question relation matrix under the answer-agreement perspective as:

$$S_q^{ag}[ij] = \frac{-2 \sum_{m \in \{0,1\}} \sum_{n \in \{0,1\}} p\left(a_i = m, a_j = n\right) \times \log \frac{p\left(a_i = m, a_j = n\right)}{p(a_i = m) p\left(a_j = n\right)}}{\sum_{m \in \{0,1\}} p(a_i = m) \log p(a_i = m) + \sum_{n \in \{0,1\}} p\left(a_j = n\right) \log p\left(a_j = n\right)} \tag{4}$$

$$\widehat{S}_q^{ag}[ij] = \begin{cases} 1, \text{if} S_q^{ag}[ij] \geqslant \lambda_q \\ 0, \text{otherwise} \end{cases} \tag{5}$$

Consequently, the set of homogeneous neighbors $N_{q_i}^{ag} = \left\{q_j | \widehat{S}_q^{ag}[ij] = 1, q_j \in Q\right\}$ of the target question q_i can be obtained, and similarly, the set of homogeneous neighbors of a certain concept c_k can also be attained by the hyperparameter λ_c.

4.3. Aggregation layer

After obtaining the set of homogeneous neighbors of questions and concepts under two semantic perspectives, to address the issue of comprehensiveness, an aggregation layer is developed. This model choice indicates our third intuition: *the relation information exists organically in questions so that those decomposed relation information should be jointly fed into question or concept embeddings*, and this choice is partly inspired by another educational work in fusing relation information [19]. Specifically, in this layer, each node in the extracted question-concept bipartite graph is updated by a two-level attention aggregation mechanism with the prior information, which aggregates the implicit signals from its homogeneous neighbors and heterogeneous neighbors.

4.3.1. Node-level attention aggregation

For the target question q_i, we utilize self-attention [21] to learn the weight of the node pair (i,j), where the node j is one of a certain type of the neighbors of the node i. First, for the target question node i, the weight of the homogeneous node pair (i,j) can be directly represented as:

$$\alpha_{ij}^{\psi} = \gamma \times \alpha_{ij}^{a} + (1-\gamma) \times \alpha_{ij}^{p} \tag{6}$$

$$\alpha_{ij}^{a} = \frac{\exp\left[\sigma\left(\mathbf{W}_{\psi}^{T}(\mathbf{q}_i || \mathbf{q}_j)\right)\right]}{\sum_{j' \in N_i^{\psi}} \exp\left[\sigma\left(\mathbf{W}_{\psi}^{T}(\mathbf{q}_i || \mathbf{q}_{j'})\right)\right]} \tag{7}$$

$$\alpha_{ij}^{p} = \frac{\exp\left(S_q^{\psi}[ij]\right)}{\sum_{j'=1}^{M} \exp\left(S_q^{\psi}[ij']\right)} \tag{8}$$

where Eq. (6) models how important homogeneous neighbor j will be for the target question node i under a certain aforementioned perspective $\psi \in \{co, ag\}$, and γ is a tunable hyperparameter to keep a trade-off between the attention score α_{ij}^{a} learned adaptively and the attention score α_{ij}^{p} computed from the prior. Furthermore, Eq. (7) and Eq. (8) model the adaptive weight coefficient and the prior weight coefficient of (i,j) respectively, where $||$ denotes the concatenate operation and $\mathbf{W}_{\psi} \in \mathbb{R}^{2d \times 1}$ is the node-level attention vector for a certain perspective ψ. As we can see from Eq. (7) and Eq. (8), α_{ij}^{a} and α_{ij}^{p} depend on their node features and statistic information respectively, and one point that needs to be highlighted is that α_{ij}^{ψ} is asymmetric which indicates that nodes in each pair may make different contributions to each other.

Then, under a certain perspective ψ, the node-level embedding $\mathbf{e}_{q_i}^{\psi}$ of the target question q_i can be aggregated by homogeneous neighbors' mapped features with the corresponding attention weights as follows:

$$\mathbf{e}_{q_i}^{\psi} = \sigma\left(\sum_{j \in N_{q_i}^{\psi}} \alpha_{ij}^{\psi} \cdot \mathbf{q}_j\right) \tag{9}$$

where without generality, $\sigma(\cdot)$ is the sigmoid function in this paper.

Thus, given the perspective set co, ag, after injecting node features and prior information into node-level attention, we can aggregate two groups of specific question node embeddings and concept node embeddings, denoted as $\left\{\mathbf{e}_{q_i}^{co}, \mathbf{e}_{q_i}^{ag}\right\}$ and $\left\{\mathbf{e}_{c_k}^{co}, \mathbf{e}_{c_k}^{ag}\right\}$.

4.3.2. Semantic-level attention aggregation

Although the target question node embedding under two semantic perspectives is obtained after node-level attention aggregation, to learn more comprehensive node embedding, it is necessary to fuse multiple semantics which can be revealed by different perspectives. Moreover, to avoid the issue of *missing semantics* (e.g., $N_{q_i}^{co} = \varnothing$ or $N_{q_i}^{ag} = \varnothing$), we

introduce direct and heterogeneous neighbors of each node (e.g., the set of direct neighbors of q_4 is $N_{q_4}^{dir} = \{c_2, c_3\}$ shown in Fig. 3) as another semantic perspective, which acts as a key booster to the achievement of semantic-level attention aggregation.

Specifically, to learn the weight of each semantic perspective of the target question node i, we first transform semantic-specific embedding through a nonlinear transformation. Then, we average all contributions from each type of neighbors of the node i. The normalized importance of each semantics for the node i, denoted as β_i^{ϕ}, is shown as follows:

$$\beta_i^{\phi} = \frac{\exp(w_i^{\phi})}{\sum_{\phi' \in \{co,ag,dir\}} w_i^{\phi'}} \tag{10}$$

$$w_i^{\phi} = \begin{cases} \frac{1}{|N_{q_i}^{\phi}|} \sum_{j \in N_{q_i}^{\phi}} \rho^T \cdot \tanh\left(\mathbf{W} \cdot \mathbf{e}_{q_j}^{\phi} + \mathbf{b}\right), \text{if} \phi \in \{co, ag\} \\ \frac{1}{|N_{q_i}^{\phi}|} \sum_{j \in N_{q_i}^{\phi}} \rho^T \cdot \tanh\left(\mathbf{W} \cdot \mathbf{c}_j + \mathbf{b}\right), \text{if} \phi \in \{dir\} \end{cases} \tag{11}$$

where $\mathbf{W} \in \mathbb{R}^{d \times d}$ is a weight matrix, $\mathbf{b} \in \mathbb{R}^{d \times 1}$ is a bias vector and $\rho \in \mathbb{R}^{d \times 1}$ is an attention vector. Note that ϕ is the element of the semantic set $\{co, ag, dir\}$. Obviously, the higher β_i^{ϕ}, the more important semantic ϕ is. With the help of learned weights, for the target question node i, we can fuse these embeddings to obtain the updated question embedding $\tilde{\mathbf{q}}_i$ as follows:

$$\tilde{\mathbf{q}}_i = \beta_i^{co} \cdot \mathbf{e}_{q_i}^{co} + \beta_i^{ag} \cdot \mathbf{e}_{q_i}^{ag} + \frac{\beta_i^{dir}}{\left|N_{q_i}^{dir}\right|} \sum_{c_k \in N_{q_i}^{dir}} \mathbf{c}_k \tag{12}$$

Similarly, for a certain concept c_k, we can also obtain its updated concept embedding $\tilde{\mathbf{c}}_k$.

4.4. Constraint layer

4.4.1. Difficulty constraint

One of the crucial goals of some downstream tasks (e.g., question recommendation or adaptive testing) is enabling questions to be suitably challenging, and it is a common notion that a good question should be neither easy nor too difficult [29–31]. Thus, in our method, we design a difficulty prediction layer to make questions possess difficulty information, which reflects our forth intuition: *difficulty information of questions plays a significant role in KT task, which, however, is not contained in the*G_{qc}. Specifically, we leverage attribute features of questions interacting with node embeddings to learn high quality question embeddings.

First, the definition of attribute features of questions is specified as the features related to the difficulty of questions, such as average response time of students and question type. In this way, for the target question q_i, its feature vector is denoted as:

$$\tilde{\mathbf{f}}_i = \sigma\left(\mathbf{W}_f[\mathbf{f}_{i1}||\mathbf{f}_{i2}||\ldots||\mathbf{f}_{iF}] + \mathbf{b}_f\right) \tag{13}$$

where $\mathbf{W}_f$ is a weight matrix, $\mathbf{b}_f$ is a bias vector, F is the number of features and $\tilde{\mathbf{f}}_i \in \mathbb{R}^d$ is the feature vector of q_i. Note that $\mathbf{f}_{ij}$ is a one-hot vector if j-th feature is categorical (e.g., question type); if j-th feature is numerical (e.g., average response time of students), then $\mathbf{f}_{ij}$ is a scalar value. Furthermore, for the target question q_i, we assume that the set of concepts related to q_i is $N_{q_i}^{dir}$ (e.g., $N_{q_1}^{dir} = \{c_1, c_2, c_3\}$) and we use the average representation of all concept embeddings in $N_{q_i}^{dir}$ as the related concept embedding of q_i, denoted as:

$$\tilde{\mathbf{c}}_{N_i} = \frac{1}{\left|N_{q_i}^{dir}\right|} \sum_{c_k \in N_{q_i}^{dir}} \tilde{\mathbf{c}}_k \tag{14}$$

Then, to recover the difficulty information from the heterogeneous graph effectively, we project the concatenated vector $\left[\tilde{\mathbf{q}}_i||\tilde{\mathbf{c}}_{N_i}||\tilde{\mathbf{f}}_i\right]$ into the final target question embedding $\hat{\mathbf{q}}_i \in \mathbb{R}^d$ by using a MLP (Multilayer Perceptron) with three layers, and further leverage a non-linear layer to map $\hat{\mathbf{q}}_i$ to a difficulty approximation: $\hat{d}_i = \sigma(\mathbf{w}_d^T \cdot \hat{\mathbf{q}}_i + b_d)$, where $\mathbf{w}_d$ and b_d are trainable parameters. Finally, following the **Definition 4**, we design the following loss function to measure the difficulty approximation error:

$$L_d = \sum_i^M \left(d_i - \hat{d}_i\right)^2 \tag{15}$$

4.4.2. Structure constraint

In the question-concept bipartite graph G_{qc}, edges exist between question vertices and concept vertices, indicating explicit signals. Inspired by the approach of modeling 1st-order proximity in LINE [20], likewise, we also attempt to model explicit relations by estimating the local proximity between question and concept vertices in the embedding space. In detail, the local proximity between question q_i and concept c_k is modeled as:

$$\hat{r}_{q_i \leftrightarrow c_k} = \sigma(\mathbf{q}_i^T \mathbf{c}_k) \tag{16}$$

To preserve structure relations, the local proximity is enforced to be close to question-concept relations in G_{qc} via a cross-entropy loss function:

$$L_s = -\sum_{q_i \in Q} \sum_{c_k \in C} [r_{q_i \leftrightarrow c_k} \log_2 \hat{r}_{q_i \leftrightarrow c_k} + (1 - r_{q_i \leftrightarrow c_k}) \log_2 (1 - \hat{r}_{q_i \leftrightarrow c_k})] \tag{17}$$

Therefore, in order to generate comprehensive question embeddings that preserve structure information and question difficulty information simultaneously, we combine these two loss functions to form a joint optimization framework, namely, we solve:

$$\min L = \min(L_d + L_s) \tag{18}$$

Once the optimization is converged, we can obtain the question embeddings $\hat{\mathbf{q}}$, which can be fed into the existing deep KT models, such as DKT, DKVMN and so on. To make our framework more readable, we present the pseudocode of pre-training question embedding in Algorithm 1.

Algorithm 1: SEEP

Input: The set of students S, the set of questions Q, the set of concepts C, the response record X of each student, and the question-concept bipartite graph G_{qc};
Output: The updated question embedding matrix $\mathbf{Q}$;
1: Initialize question embedding matrix $\mathbf{Q}$, concept embedding matrix $\mathbf{C}$ and other weight matrices;
2: **for** $q_i \in Q$ **do**
3: obtain the homogeneous set $N_{q_i}^{co}$ of q_i by Eq. (2) and Eq. (3);
4: obtain the homogeneous set $N_{q_i}^{ag}$ of q_i by Eq. (4) and Eq. (5);
5: **end for**
6: Obtain the homogeneous sets $N_{c_k}^{co}$ and $N_{c_k}^{ag}$ of each concept c_k in a similar way;
7: **for** $q_i \in Q$ **do**
8: // Node-level Attention Aggregation
9: Obtain two groups of question node embedding $\mathbf{e}_{q_i}^{co}$ and $\mathbf{e}_{q_i}^{ag}$ for q_i by Eq. (6)–(9);
10: **end for**
11: Two groups of concept node embedding $\mathbf{e}_{c_k}^{co}$ and $\mathbf{e}_{c_k}^{ag}$ for c_k can be obtained in a similar way;
12: **for** $q_i \in Q$ **do**
13: // Semantic-level Attention Aggregation
14: Obtain the updated question embedding $\tilde{\mathbf{q}}_i$ by Eq. (10)–(12);
15: **end for**
16: Compute and optimize the loss function by Eq. (13)–(18)
17: **return** The updated question embedding matrix $\mathbf{Q}$;

4.5. Model analysis

In this subsection, we further analyze the time complexity of the main components in each module in our SEEP and conduct some analysis concerning SEEP's relation with some existing methods (i.e., RCD and PEBG).

4.5.1. Complexity analysis

In the decomposition layer, the computation mainly comes from the computing of matrices $\hat{\mathbf{S}}_q^{co}, \hat{\mathbf{S}}_q^{ag}, \hat{\mathbf{S}}_c^{co}$ and $\hat{\mathbf{S}}_c^{ag}$ (Eq. (2)–(5)). The time complexity is $O\left(2\left(M^2 + K^2\right)\right)$, where M and K denote the number of questions and concepts respectively. In the aggregation layer, the time complexity mainly depends on the computation of node-level and semantic-level attention of each node (Eq. (6)–(12)), which incurs $O\left((M+K)\left(2d + d^2\right)\right)$. Here, d is the embedding size. In the constraint layer, the time complexities of difficulty constraint and structure constraint are $O(MdF)$ and $O(MK)$, where F indicates the number of features.

4.5.2. Relation with RCD

RCD [19] uniformly models the interactive and structural relations via a multi-layer student-question-concept relation map to estimate students' hidden knowledge states. Our SEEP method also uses the student-question-concept interaction but distinguishes from RCD in several aspects: (1) RCD constructs the student-question-concept relation map as a uniform heterogeneous graph and learns three types of node embeddings for cognitive diagnosis. However, in SEEP method, we extract the question-concept interaction as a heterogeneous graph to learn two types of node embeddings (i.e., question nodes and concept nodes) and leverage the response information in student-question interaction to construct the relations between questions (concepts); (2) RCD models the edges between student nodes and question nodes by the set of interactions from training data, where students' exercising response data (i.e., answer correctly or not) are not revealed in fact. In SEEP, students' exercising response data are utilized to construct the relations between questions (concepts) in a more fine-grained manner.

4.5.3. Relation with PEBG

PEBG leverages the side information including question difficulties together with three kinds of relations: explicit question-concept relations, implicit question similarity and concept similarity. The SEEP method differs from PEBG in the following perspectives: (1) PEBG solely models the question-concept heterogeneous graph, while SEEP models the student-question-concept interaction; (2) PEBG models the implicit question similarity and concept similarity by simple folding signals, while SEEP models not only explicit question-concept relations but also implicit question similarity and concept similarity derived from student-question response data.

5. Experiments

In this section, we conduct extensive experiments to evaluate the performance of our proposed SEEP method. Specifically, we first describe the datasets and introduce the experimental setup. Then we demonstrate the effectiveness and the semantic-enhanced property of our method by answering the following research questions:

- RQ1: Comparing with the state-of-the-art question embedding method without text information (i.e., PEBG), how does our proposed SEEP method perform in terms of accuracy?
- RQ2: How about the KT models' performance when SEEP blends relation information and graph structural information into current models?
- RQ3: How do the hyperparameters (e.g., $\lambda_q, \lambda_c, \gamma$) tuned in the decomposition layer and the aggregation layer affect SEEP effectiveness?
- RQ4: How about the interpretation of question difficulty recovered by SEEP method?

5.1. Dataset fescription

Two real-world datasets have been utilized for our experiments. One is ASSISTment09[1] (ASSISTments2009–2010 skill builder), an open dataset collected by the ASSISTments online tutoring systems, which is widely used by researchers for purposes such as KT and student modeling. Besides, we choose the public corrected version which mitigates the duplicated data problem proposed by previous research [18]. The other dataset is Junyi[2] supplied by an E-learning platform established in 2012 on the basis of the open-source code released by Khan Academy, which is collected by JunyiAcademy in 2015 [19]. Table 3 summarizes basic statistics of these two preprocessed datasets.

5.2. Experimental setup

5.2.1. Implementation settings

We set the embedding size $d = 128$ and $\gamma = 0.5$. During the relation map exploitation, we set $\lambda_q = \lambda_c = 0.5$ (reported in Fig. 4). Besides, we implement the initialization of the whole parameters depending on *Xavier* [32], which fills the weights with random values sampled from $N \sim \left(0, std^2\right)$, where $std = \sqrt{2/(n_{in} + n_{out})}$, n_{in} is the number of neurons feeding into the weights and n_{out} is the number of neurons which the results are sent to. Meanwhile, we train the SEEP with *Adam* [33] and use learning rate of 0.001 and mini-batch size of 64.

We use PyTorch to implement our method and conduct our experiments on a server with Quadro RTX 6000 GPU with 24G of video memory.

5.2.2. Baselines and evaluation metrics

To show the improvement of our method to the mainstream KT models and verify the effectiveness of our proposed method, we compare it with some baselines. Specifically, we first applied it on several kinds of KT models like memory-

[1] https://sites.google.com/site/assistmentsdata/home/assistment-2009-2010-data/skill-builder-data-2009-2010
[2] https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=1198

Table 3
The statistics of the datasets.

Statistics	ASSISTment09	Junyi
#Students	4,003	14,605
#Questions	3,215	830
#Concepts	123	784
#Response records	50,551	20,803,598
#Concepts per question	2.17	1.12
#Std of concepts per question	0.47	0.26
#Questions per concept	172.69	8.23
#Std of questions per concept	169.58	4.18

based models (DKT, DKVMN, LPKT), transformer-based models (SAKT, AKT) and graph-based models (GIKT), and then we compare it with the state-of-the-art question embedding method without text information PEBG. The details are illustrated as follows:

- *DKT* [5] is an implementation using a LSTM which leverages a series of one-hot representations of question and trains a unified network for predicting students' next interaction on certain question.
- *DKVMN* [6] uses a key-value memory network to store underlying concept representations and states.
- *LPKT* [7] monitors students' knowledge state through directly modeling their learning process to achieve the KT task.
- *SAKT* [8] is a recently proposed deep-learning method based on self-attention mechanism to capture the relation between interactions.
- *AKT* [10] is an transformer-based neural network model and attention weights are computed by a distance-aware exponential decay with a global decay rate, where we use two variants of AKT including AKT-NR (without Rasch model-based embeddings) and AKT-R (with Rasch model-based embeddings).
- *GIKT* [12] employs high-order question-concept relation graphs into question and skill representations for knowledge tracing.
- *PEBG* [16] formulates the question-concept relations as a bipartite graph and introduces a product layer to learn low-dimensional question embeddings for knowledge tracing.
- *PERM* [22] utilizes the relation map to pre-train question embeddings for improving knowledge tracing.

Here we do not include RKT [9] and GKT [11] mentioned in Section 2 because: 1) the implementation of RKT requires the text information of questions; 2) a question utilized in GKT solely contains a single concept. In order to ensure the fairness, all models were tuned to have the best performance. For each dataset, the training process is repeated five times, and we report the average test results.

Moreover, we select evaluation metrics from both regression perspective and classification perspective. From the regression perspective, we select Root Mean Square Error (RMSE, the lower the better) [34] to quantify distances between predicted scores (i.e., continuous variable ranges from 0 to 1) and actual ones. From the classification perspective, we argue that students' response logs answered incorrectly or correctly can be represented as 0 and 1 respectively. Thereby, we select Accuracy (ACC, the higher the better) [18] and Area Under the ROC Curve (AUC, the higher the better) [35] for model evaluation.

5.3. Performance comparison (RQ1)

To answer the RQ1, we test our method SEEP on the aforementioned models and compare their performance with both PEBG + DKT and PEBG + DKVMN following the literature [16].

Table 4 and 5 report the overall performance of all methods on the student performance prediction task and the best scores are denoted in bold. From the results, we summarize several important observations. Firstly, almost all of the pre-training-based SEEP + methods perform better than baselines at all data sparsities, which may indicate that taking the student-question-concept interaction with the issue of co-existing heterogeneity and comprehensiveness into account for question embedding can achieve outstanding performance on the downstream KT task. Secondly, SEEP + DKT and SEEP + DKVMN consistently perform better than PEBG + and PERM+. This has two possible implications: (1) PEBG fails to take students' response logs into consideration; (2) PEBG learns question embeddings and concept embeddings by constructing an embarrassingly simple folded bipartite graph, which may introduce noisy signals into both question and concept embeddings; (3) PERM does not take question difficulty information into account. Lastly, almost all SEEP + methods are superior to baselines, probably because SEEP has the ability to decompose the implicit information in the interaction and further fusing the information of questions and concepts under different decomposed perspective semantics, which enhances relation information for questions and concepts.

Table 4
Experimental results of student performance prediction on ASSISTment09.

Train/Test ratio		50%/50%			60%/40%			70%/30%			80%/20%		
Methods		RMSE	ACC	AUC	RMSE	ACC	AUC	RMSE	ACC	AUC	RMSE	ACC	AUC
Baseline	DKT	0.471	0.631	0.683	0.467	0.638	0.687	0.464	0.714	0.737	0.458	0.731	0.740
	DKVMN	0.468	0.648	0.692	0.453	0.647	0.694	0.457	0.721	0.746	0.441	0.747	0.753
	SAKT	0.463	0.651	0.708	0.449	0.659	0.702	0.441	0.734	0.751	0.435	0.758	0.762
	AKT-NR	0.458	0.667	0.718	0.442	0.664	0.716	0.438	0.742	0.768	0.428	0.764	0.778
	AKT-R	0.439	0.674	0.722	0.431	0.682	0.725	0.423	0.764	0.771	0.402	0.776	0.801
	GIKT	0.441	0.672	0.721	0.436	0.678	0.722	0.429	0.759	0.776	0.411	0.771	0.794
	LPKT	0.432	0.678	0.727	0.428	0.688	0.731	0.417	0.721	0.779	0.398	0.782	0.813
Pre-training based	PEBG + DKT	0.452	0.658	0.709	0.441	0.673	0.711	0.435	0.747	0.764	0.424	0.768	0.781
	PEBG + DKVMN	0.446	0.662	0.715	0.438	0.679	0.719	0.427	0.758	0.778	0.412	0.774	0.797
	PERM + DKT	0.432	0.673	0.724	0.431	0.687	0.721	0.422	0.764	0.784	0.408	0.783	0.803
	PERM + DKVMN	0.421	0.681	0.728	0.427	0.694	0.729	0.413	0.771	0.799	0.401	0.794	0.814
	SEEP + DKT	0.429	0.684	0.733	0.424	0.693	0.726	0.418	0.769	0.791	0.402	0.788	0.809
	SEEP + DKVMN	0.417	0.687	0.735	0.421	0.703	0.737	0.408	0.776	0.802	0.396	0.798	0.821
	SEEP + SAKT	0.419	0.687	0.731	0.414	0.709	0.739	0.407	0.779	0.808	0.394	0.807	0.827
	SEEP + AKT-NR	0.408	0.698	0.736	0.412	0.718	0.748	0.398	0.788	0.814	0.388	0.818	0.839
	SEEP + GIKT	0.399	0.708	0.744	0.396	0.731	0.756	0.392	0.813	0.823	0.383	0.836	0.843
	SEEP + LPKT	**0.382**	**0.734**	**0.751**	**0.381**	**0.743**	**0.764**	**0.371**	**0.821**	**0.832**	**0.367**	**0.841**	**0.854**

Table 5
Experimental results of student performance prediction on Junyi.

Train/Test ratio		50%/50%			60%/40%			70%/30%			80%/20%		
Methods		RMSE	ACC	AUC	RMSE	ACC	AUC	RMSE	ACC	AUC	RMSE	ACC	AUC
Baseline	DKT	0.496	0.599	0.663	0.493	0.627	0.661	0.467	0.694	0.703	0.462	0.706	0.727
	DKVMN	0.488	0.607	0.671	0.482	0.632	0.673	0.453	0.706	0.718	0.451	0.718	0.736
	SAKT	0.481	0.618	0.683	0.474	0.64	0.684	0.449	0.71	0.726	0.443	0.725	0.745
	AKT-NR	0.473	0.624	0.691	0.468	0.648	0.691	0.445	0.718	0.732	0.439	0.733	0.753
	AKT-R	0.469	0.637	0.699	0.461	0.653	0.702	0.439	0.726	0.743	0.433	0.745	0.768
	GIKT	0.441	0.672	0.721	0.436	0.678	0.722	0.429	0.759	0.776	0.411	0.771	0.794
	LPKT	0.434	0.681	0.733	0.428	0.687	0.731	0.421	0.764	0.781	0.413	0.779	0.803
Pre-training based	PEBG + DKT	0.470	0.637	0.694	0.462	0.652	0.697	0.441	0.717	0.738	0.433	0.738	0.758
	PEBG + DKVMN	0.464	0.644	0.701	0.457	0.666	0.708	0.433	0.722	0.747	0.429	0.749	0.769
	PERM + DKT	0.461	0.654	0.709	0.451	0.675	0.713	0.437	0.728	0.754	0.425	0.757	0.773
	PERM + DKVMN	0.453	0.668	0.716	0.442	0.683	0.721	0.429	0.731	0.761	0.419	0.761	0.781
	SEEP + DKT	0.458	0.661	0.713	0.448	0.681	0.719	0.432	0.736	0.762	0.419	0.763	0.781
	SEEP + DKVMN	0.447	0.671	0.724	0.436	0.689	0.726	0.421	0.735	0.766	0.413	0.767	0.785
	SEEP + SAKT	0.445	0.673	0.723	0.439	0.691	0.728	0.425	0.739	0.768	0.412	0.769	0.787
	SEEP + AKT-NR	0.439	0.687	0.728	0.433	0.697	0.736	0.418	0.748	0.776	0.404	0.776	0.794
	SEEP + GIKT	0.431	0.693	0.737	0.424	0.708	0.749	**0.401**	**0.759**	**0.787**	**0.397**	**0.784**	**0.807**
	SEEP + LPKT	**0.428**	**0.699**	**0.742**	**0.416**	**0.712**	**0.754**	0.413	0.743	0.781	0.401	0.779	0.802

5.4. Ablation study (RQ2)

To answer RQ2, we conduct an ablation study to investigate the effectiveness of three important components of our SEEP method on ASSISTment09 and Junyi: (1) prior weights; (2) the decomposition layer; (3) the aggregation layer. We further set three comparative settings, and their performances have been shown in Fig. 5 and 6. The details of the three settings are listed below:

- SEEP + GIKT/P does not consider the prior weights in the node-level aggregation with $\gamma = 1$ in the Eq. (6).
- SEEP + GIKT/DL does not consider the decomposition layer, where we leverage a folded bipartite graph to select homogeneous neighbors for questions or concepts like PEBG.
- SEEP + GIKT/AL does not consider the aggregation layer, where we choose another aggregation strategy following the literature [36] to aggregate relation information from both homogeneous and heterogeneous neighbors for the target question or concept node.
- SEEP + GIKT/D does not consider the difficulty constraint by removing Eq. (15) from Eq. (18).
- SEEP + GIKT/S does not consider the structure constraint by removing Eq. (17) from Eq. (18).

To be fair, except for the changes mentioned above, the other parts of aforementioned methods and implementation settings remain identical. From Fig. 5 and 6 we can find that: (1) SEEP + GIKT performs the best, indicating the effectiveness of

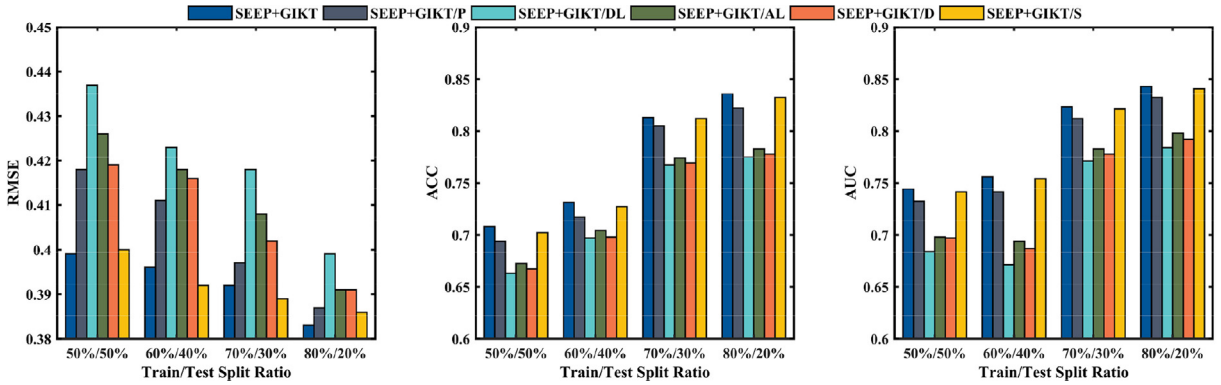


Fig. 5. Performance comparison of ablation study on ASSISTment09.

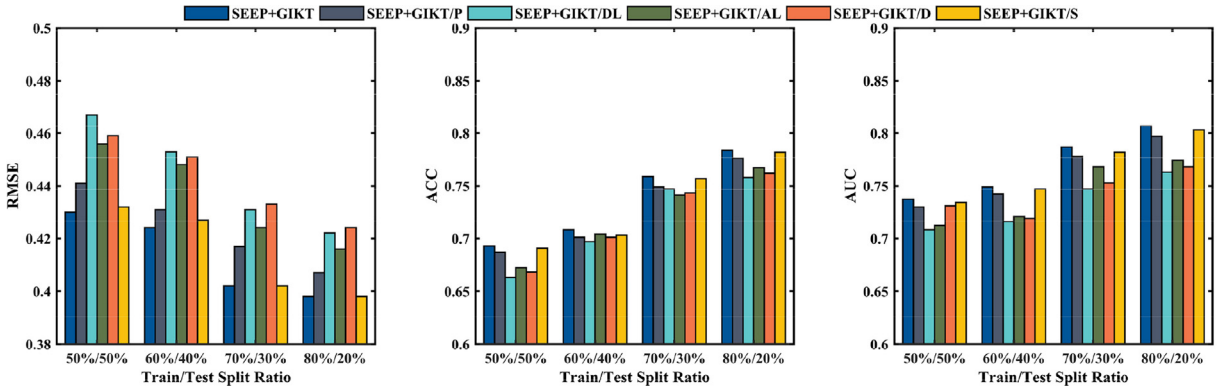


Fig. 6. Performance comparison of ablation study on Junyi.

different components of our SEEP method; (2) the performances decline slightly when removing prior weights in the node-level aggregation, which can be postulated that SEEP is not sensitive to prior weights or SEEP prefers to adaptively learned weights rather than prior weights (which will be discussed in Section 5.5); (3) removing the decomposition layer hurts the performances badly and utilizing the folded bipartite graph also has lower performances, which confirms that simply treating question-concept interaction as a folded bipartite graph is problematic and demonstrates the rationality of decomposing implicit information explicitly from the heterogeneous graph (i.e., the student-question-concept interaction); (4) replacing the aggregation layer with a light GCN layer, the performances also show a shrinking trend, which indicates that two-level aggregation mechanism with the prior information can take effects in enhancing the representation ability of question embeddings based on the decomposition layer; (5) the performances degrade substantially by removing the difficulty constraint, where the reason for this result may be that the difficulty is a crucial property for one question; (6) the performances decline moderately after removing the structure constraint, which may hint at the potential of labeled edge signals between question nodes and concept nodes in G_{qc}.

5.5. Efficiency analysis

We present the computation time of our SEEP with two pre-training baseline methods in Table 6. The results demonstrate the high computation efficiency of our model. In terms of the model size, SEEP has fewer parameters than PEBG models but more than PERM. In the training phase, our model runs faster than PEBG except for PERM. By further considering the prediction accuracy (see Table 4 and 5), our model shows superior ability in balancing predictive performances and time consumption as well as parameter settings.

5.6. Sensitivity evaluation (RQ3)

To answer RQ3, in this section, we investigate the effect of hyperparameters of SEEP method with GIKT model according to the performance (AUC) of the student performance prediction task on ASSISTment09. More specifically, we report the sensitivity analysis of three significant hyperparameters in SEEP, including the sparsity factor of homogeneous neighbors of question q (λ_q), the sparsity factor of homogeneous neighbors of concept c (λ_c) and the balance factor between prior weights and learned weighs (γ). The detailed results are presented in Fig. 7.

Table 6
Comparisons of computation time. The training time is the time cost per epoch in the training phase, and the predicting time is the total time cost on the validation set.

Dataset		SEEP	PEBG	PERM
ASSISTment09	Training time (s/epoch)	49.82	118.06	32.27
	Predicting time (s)	5.22	18.7	4.06
Junyi	Training time (s/epoch)	88.41	306.66	56.18
	Predicting time (s)	11.84	45.13	6.03

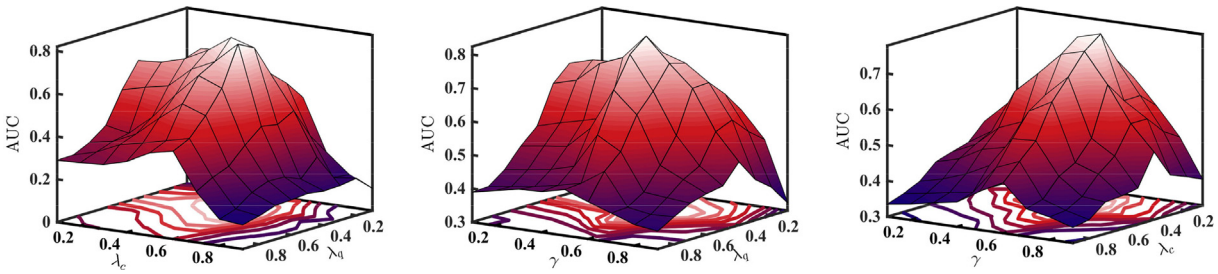


Fig. 7. Hyperparameter sensitivity analysis of the SEEP + GIKT model on ASSISTment09.

- **The sparsity factor of homogeneous neighbors of question** q **(**λ_q**).** We can find that the performance (i.e., AUC) is improved by changing the sparsity factor λ_q from 0.3 to 0.6. However, as λ_q increases further, the performance shrinks dramatically. The main reason is that larger value of λ_q means homogeneous neighbors of question q are obliged to be more similar to the question q, which has remained stubbornly beyond the reach of most homogeneous neighbors.
- **The sparsity factor of homogeneous neighbors of question** q **(**λ_c**).** For the sparsity factor λ_c, the impact shows similar patterns. As λ_c increases, the performance raises first, and then reaches the best performance when $\lambda_c = 0.6$. Afterwards, the performance begins to go considerably down. The sparsity factor λ_c with a bigger value can generally encode more meaningful relation information for both question and concept embeddings, whereas the sparsity factor λ_c with too bigger value may introduce additional redundancies and lead to over-fitting.
- **The balance factor between prior weights and learned weighs (**γ**).** Intuitively, a larger balance factor γ indicates that our SEEP method prefers weights learned adaptively rather than the prior when performing node-level aggregation. In effect, changing trends of performance demonstrate the former statement indeed, and it is more probable that the prior existed on some datasets is redundant to some extent. However, we find that the prior still serves as a vitamin to question embeddings in specific datasets, that is, even in small quantities could achieve comparable performance, while its deficiency would hinder the results.

5.7. Case study (RQ4)

As question embeddings can reflect difficulty information, we explicitly label each interaction with 'correctly' or 'incorrectly' via the true interaction records. Then, we leverage the t-SNE method [27] to reduce the dimension of question embeddings to two-dimension space to represent question difficulty information. We randomly select 1000 interactions on ASSISTment09 dataset for visualization. As shown in Fig. 8(a) and (b), those interactions in the same color have the same

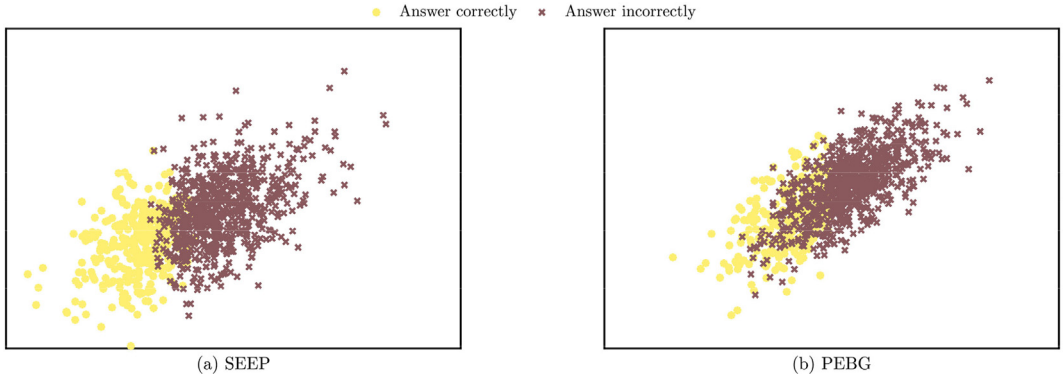


Fig. 8. The visualization of the learned question embeddings by using SEEP and PEBG on ASSISTment09.

label. In our SEEP method, we can see that interactions answered correctly are mainly clustered on the left side of the figure, and interactions answered incorrectly are mainly clustered on the right side of the figure. In particular, the overlapping area between interactions answered correctly and interactions answered incorrectly is small, which means that bias between the question difficulty information recovered by SEEP and the true students' performance data is small. In the contrast, the overlapping area in PEBG is relatively large, and it is probable that introducing a number of extra embarrassingly simple folded signals between questions (concepts) produces noises.

6. Conclusions and future work

In this paper, we present a novel SEEP method, concentrating on exploting semantic relation maps and further fusing the embeddings of question and concept under two perspective semantics to obtain the final question embedding for improving performances of KT methods. Extensive experiments conducted on two real-world datasets show SEEP has higher expressive power that enables KT methods to effectively predict students' performance. Avenues of future work include: (1) generalizing existing question embeddings to more fine-grained embeddings (e.g., correct or incorrect question embeddings) and (2) fusing more psychometric or signed network theories into question embedding methods.

CRediT authorship contribution statement

Wentao Wang: Conceptualization, Methodology, Software, Writing - original draft. **Huifang Ma:** Supervision, Writing - review & editing, Funding acquisition. **Yan Zhao:** Visualization, Investigation. **Fanyi Yang:** Supervision, Investigation. **Liang Chang:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Huifang Ma reports financial support was provided by the National Natural Science Foundation of China.

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